Why Cognitive Behavioral Therapy Can Fail: A Critical Meta-Review Across Conditions

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Protocol version 0.2; developing critical review

Abstract

Cognitive behavioral therapy (CBT) can benefit patients while leaving substantial nonresponse and uncertainty about its specific mechanisms (Loerinc et al., 2015; Carpenter et al., 2018). Evidence for a multicomponent treatment does not identify cognitive restructuring as the necessary source of benefit (Jacobson et al., 1996). We develop a critical meta-review of reasons CBT can fail across conditions, including anxiety, depression and other clinical applications. The protocol separates nonresponse, comparator choice, bias, component mechanisms, executive demands, durability and treatment burden. Adult attention-deficit/hyperactivity disorder and baseline executive function form one case study. The current report provides a cited seed synthesis, a preliminary screen of 21 case-study records, reproducible ADHD search receipts and preserved author drafts. Cross-condition evidence supports benefits while showing that interpretation depends on the comparator and condition (Cuijpers et al., 2025). A defensible critique must identify the treatment, outcome and population for which a limitation is supported, rather than infer universal ineffectiveness from an uncertain mechanism.

1 Introduction

Average symptom improvement does not settle whether cognitive behavioral therapy (CBT) adequately serves the people offered it. A systematic review of anxiety treatment response found that conclusions depend on how response is defined and which analysis population is used (Loerinc et al., 2015). A cross-condition synthesis found benefits that depended on the control condition, alongside limitations in trial quality and coverage (Cuijpers et al., 2025). These findings motivate a critical review of nonresponse, comparative value and the scope of claims made for CBT.

The author's archived blog and Google Docs draft begin with a concern about reasoning through distress: cognitive restructuring may demand capacities that stress makes harder to use. Stress-related prefrontal impairment provides a biological rationale for investigating this concern, but does not establish treatment failure (Arnsten, 2009). The original drafts also ask whether exposure, behavioral practice and support explain benefits attributed to cognitive restructuring. A depression component trial did not identify an advantage of the full cognitive-therapy package over its component treatments, providing a direct but bounded challenge to the necessity of the complete package (Jacobson et al., 1996).

The clinical evidence also constrains the critique. Placebo-controlled anxiety trials report benefits of CBT, so a universal claim that CBT does not work is inconsistent with this retrieved evidence (Carpenter et al., 2018). For depression, a large synthesis found benefit relative to controls but a small advantage over other psychotherapies that was not robust across most sensitivity analyses

(Cuijpers et al., 2023). These are different questions: treatment benefit, superiority over alternatives and support for a particular mechanism should not be collapsed into one verdict.

We therefore organize the review around distinct reasons for failure or overstatement across clinical conditions. The scope includes incomplete response, comparator-sensitive effects, study bias, component attribution, executive demands, long-term functioning, acceptability and harms. The current evidence map is a targeted starting synthesis; comprehensive cross-condition screening and quantitative extraction remain unfinished. We preserve the author's skeptical argument and require each empirical claim to survive contradictory evidence and condition-specific assessment.

Adult attention-deficit/hyperactivity disorder (ADHD) is one case study within this broader review. Existing reviews and trials describe benefits and limitations of CBT in adult ADHD, including executive-function and functioning outcomes (Lopez et al., 2018; Safren et al., 2010; Liu et al., 2026; Lopez-Pinar et al., 2026). A baseline predictor of outcome is distinct from a moderator of the randomized treatment contrast, and improvement in executive function is not itself evidence of baseline moderation (Kraemer et al., 2002). Related older-adult studies of executive function and flexibility provide indirect evidence whose population and analyses must remain explicit (Mohlman and Gorman, 2005; Johnco et al., 2014).

This manuscript contributes the amended protocol, a cited critical evidence map and a preliminary screen of the 21 records in the ADHD executive-function search slice. The search baseline remains reproducible and is not relabeled as coverage of all CBT research. Systematic selection and search reporting will follow the Preferred Reporting Items for Systematic Reviews and Meta-Analyses framework and its search extension (Page et al., 2021; Rethlefsen et al., 2021).

2 Methods and protocol

Protocol version 0.2, dated October 21, 2026, restores the author's cross-condition scope and has not been registered. The full operational amendment is maintained in `protocol.md`. The primary stream includes systematic reviews and meta-analyses of CBT in adults across clinical conditions, with each condition and comparator retained separately. Supplementary streams examine component trials, moderators, mechanisms, harms and implementation. Outcomes include symptoms, response, remission, functioning, quality of life, durability, deterioration, dropout and burden. The initial ten-query search and the executive-function screening exercise are an ADHD case study, not the primary review's population restriction.

For that case study, eligible populations are adults with diagnosed ADHD, and mixed samples require separately extractable findings. Intervention components and delivery formats are recorded separately. Comparators are classified as passive, active psychological, or medication-only for adjunctive treatment. Across conditions, the meta-review includes systematic reviews with explicit search and selection methods. The ADHD baseline-executive-function stream requires pretreatment executive-function measurement and a later outcome association or treatment interaction. Single-arm associations are prognostic evidence, not causal treatment-effect moderation. Post-treatment executive-function scores alone do not enter the baseline moderator synthesis. We will record whether measures are performance-based or self-reported, as well as diagnosis, medication, comorbidity, outcome overlap, missingness and prespecification.

Reviews will be appraised using AMSTAR 2, with domain judgments retained (Shea et al., 2017). The original-study risk-of-bias instrument will be fixed in a protocol amendment before extraction. Review overlap will be mapped at the primary-study level; overlapping meta-analyses will not be pooled as independent estimates. Quantitative moderation synthesis remains conditional on compatible constructs, comparisons and interaction estimands, with a statistical analysis amendment

required before fitting.

3 Search strategy

The retained ADHD case study uses ten overlapping search slices covering reviews, trials, executive-function associations, broader executive-function terms, specific cognitive processes, medication, components and comparators, adherence, functioning and anxiety. The complete Europe PMC, PubMed and PsycINFO translations are versioned in `search/queries.json`; Europe PMC strings are reproduced in the protocol. The baseline publication-date cutoff is October 21, 2026. Only Europe PMC has been executed in this initialization. PubMed and PsycINFO remain planned; the Ovid PsycINFO translations require institutional syntax validation.

The standard-library Python script retrieves match counts, preserves request queries and response hashes, and appends dated rows to a comma-separated values (CSV) log. Each response includes only one record identifier, so this operation is not a complete screening export. A failed request is logged with a blank count and an error, never as zero. The historical notes of 27 reviews and 9 executive-function/moderator/outcome hits lack their exact search strings and dates and remain unverified. Current queries are not tuned to reproduce those numbers. Complete record exports, deduplication, reference checking and search sensitivity checks are required before eligibility or completeness claims, consistent with search-reporting guidance (Rethlefsen et al., 2021).

The broader evidence map uses targeted retrieval of identified seed sources from the author drafts and related syntheses. A systematic cross-condition search is pending; the case-study counts cannot be interpreted as its results.

4 Screening and extraction plan

Two independent reviewers will pilot the eligibility rules and screen titles/abstracts and full texts, with an adjudicator resolving remaining disagreements. Reviewers have not yet been assigned, and no independent screening is claimed in this draft. The 21-record ADHD case-study screen is explicitly preliminary and single-agent; unresolved full texts remain open. Full-text exclusions will receive explicit reasons; unavailable texts will remain awaiting retrieval. Companion reports will be linked to underlying studies, and duplicates across databases and reviews will be tracked.

Extraction will record outcome-specific denominators, intervention components, executive-function timing and instrument, comparison group, rater, follow-up, estimate, confidence interval, reported significance test, covariates, missingness and prespecification. One-arm associations, between-arm interactions and changes in executive function will be labeled separately. The selection flow will be populated only after screening, following PRISMA reporting guidance (Page et al., 2021).

5 Results: critical evidence map and ADHD case study

All ten Europe PMC searches completed successfully on October 21, 2026, with the publication-date cutoff fixed to that day. Table 1 reports the endpoint match counts; these are not screened studies and overlap across slices. Counts are database retrieval totals, not inferential estimates, so confidence intervals and significance tests do not apply. No pooled clinical estimate or completed independent screening total is claimed.

The source archive contains the author's blog and prior 29-page critical review. The current manuscript has sixteen independently identified bibliographic records, each linked to a scope-limited

Slice	Search focus	Matches
S01	Reviews and meta-analyses	40
S02	Randomized trials	73
S03	Executive function and predictors or moderators	21
S04	Executive function sensitivity search	24
S05	Specific executive processes	10
S06	Medication and adjunctive treatment	63
S07	Active comparators and treatment components	22
S08	Adherence and feasibility	23
S09	Functioning and follow-up	71
S10	Comorbid anxiety	54

Table 1: Europe PMC baseline, protocol 0.1, run `baseline-10-21-2026`. All 10 requests succeeded on their first attempt. The complete queries and receipts are retained in the repository.

claim ledger. These initialization outputs establish provenance and a review plan, not a completed clinical evidence synthesis.

5.1 Cross-condition seed synthesis

The cited evidence map separates nonresponse, comparator dependence, component attribution, comparative value, bias, executive demands, durability and burden. It is a targeted synthesis, not an exhaustive systematic result. Detailed source links and remaining gaps are recorded in `evidence-map.md`.

5.2 Preliminary screening of 21 executive-function records

A full export of the ADHD executive-function/predictor slice returned 21 records on October 21, 2026. Single-agent screening confirmed no direct baseline executive-function predictor or treatment-moderator test in the retrieved material. Sixteen records were classified as not direct tests, while five candidate full texts remain unresolved. These are record-level descriptive counts, not unique-trial totals or inferential estimates; confidence intervals and significance tests do not apply. The full titles, identifiers, evidence levels, reasons and source locations appear in `data/s03-screening.csv` and its Markdown companion.

The unresolved candidates have PubMed identifiers 39798484, 41962667, 41220055, 33615398 and 28866911. The emotion-regulation profile report (41962667) is indirect: its grouping variables are not baseline executive-function measures, and its full outcome model remains to be assessed. The other abstracts mainly describe executive-function outcomes or mediation rather than direct baseline prediction. Absence from an abstract does not establish absence from the full study.

Two preprint records yielded byte-identical primary PDFs despite differing indexed descriptions; another has an apparently overlapping sample and a different allocation description. They are flagged for provenance clarification and cannot be treated as independent replications. Protocols and companion publications are also linked in the ledger. This provisional screen is evidence of a retrieval and classification gap, not evidence that baseline executive function never matters or that CBT fails generally.

6 Discussion: reasons for failure and limits of the critique

6.1 Nonresponse, comparators and distinctive value

A review focused on failure must examine individual response and functioning rather than treating a favorable average as sufficient. Anxiety response estimates depend on definitions and analysis populations (Loerinc et al., 2015). Comparator-dependent results across conditions make the choice between waitlist, usual care and credible alternatives central to interpretation (Cuijpers et al., 2025). For depression, limited robustness of superiority over other psychotherapies supports caution about claims of unique comparative advantage, while the same synthesis reports benefit against controls and at follow-up (Cuijpers et al., 2023).

6.2 Component explanations and executive demands

Package efficacy does not establish that changing thoughts through deliberate reasoning is necessary for benefit. The depression component trial challenges that inference for its tested comparisons, but a nonsignificant advantage does not prove equivalence or universal dispensability of cognitive work (Jacobson et al., 1996). Likewise, stress-related prefrontal impairment motivates an executive-demand hypothesis without directly testing CBT outcomes (Arnsten, 2009). The baseline predictor, treatment moderator and post-treatment outcome remain separate evidence categories (Kraemer et al., 2002). The preliminary ADHD screen cannot close this question while candidate full texts remain unresolved.

6.3 Bias, durability, burden and harms

Completer analyses can differ from intention-to-treat estimates in anxiety trials, so missingness and analysis populations require explicit extraction (Carpenter et al., 2018). The cross-condition synthesis identifies limitations in study quality, heterogeneity and publication bias, and its focus on post-treatment outcomes leaves longer-term questions open (Cuijpers et al., 2025). Dropout may indicate acceptability problems, but does not identify harm or its cause. Dedicated searches for deterioration, adverse events, opportunity costs and real-world access are still needed; these are research questions rather than established universal failures.

6.4 What the current review can claim

The retrieved evidence justifies specific criticisms and a broader research program; it also includes evidence of benefit that the argument must accommodate (Carpenter et al., 2018; Cuijpers et al., 2023, 2025). The prior author drafts are preserved as the source of the skeptical argument, not substituted for a completed systematic review. The current seed map has incomplete coverage, and the case-study screening is preliminary and conducted by one agent. No pooled cross-condition treatment effect, universal efficacy verdict or individual treatment recommendation follows from this draft.

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A Source archive and reproducibility

The repository preserves the author's original blog Markdown, a Google document text export, and the 29-page deep-research Draft 2 PDF with extracted text. Source identifiers and SHA-256 hashes are recorded in `sources/manifest.json`. The imported review motivates the broader anxiety and cognitive-restructuring question and identifies itself as a critical theoretical review rather than a completed systematic review. Its claim of approximately 300 identifier-verified sources is not independently certified here; the underlying ledger was not retrieved. Its journal-style submission footer is not evidence of an actual submission.

The current bibliography uses separately retrieved Europe PMC metadata, with a claim-and-scope ledger in `data/citation-evidence.csv`. Metadata and abstract checks establish source identity and support the limited statements used here; they do not replace full-text assessment for the review. Run `make test` for deterministic checks and `make paper` to compile this manuscript with latexmk and BibTeX. Search reruns use a new run label and retain earlier receipts. The public repository is `https://github.com/brando90/meta-review-cbt`.